

**VIT**Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAIReg. Number: **Continuous Assessment Test(CAT) – II - SEPTEMBER 2026**

Programme	: B.Tech	Semester	:	Fall 2026-27
Course Code & Course Title	: BECE355L - Advanced Cloud Computing	Class Number	:	CH2026270102409 CH2026270102420 CH2026270102402 CH2026270102412 CH2026270102415 CH2026270102422
Faculty	: Dr. Arivarasi A Dr. Raju Patel Dr. Rohith G Dr. Upender P Dr Prasanna Bharathi Dr Sritama Roy	Slot	:	C1+TC1
Duration	: 90 Minutes	Max. Mark	:	50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write any other information.

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	Blooms Taxonomy Level
1		An e-commerce company wants to analyze user behaviour on their website in real-time. They need to capture user clicks, page views, and other events to personalize the user experience, recommend products, and monitor website performance. Which AWS service would you choose and why? The data must also be archived for future analysis. Draw and design a cost-effective and scalable architecture for the given scenario. [10 Marks]	10	2	3
2		A company has migrated its student-management application to AWS. Developers, testers, and administrators all work in the same AWS environment using highly privileged credentials, and a developer recently deleted a production resource by mistake. As an AWS cloud engineer, Explain a service based access-control solution that: [8 Marks] a. Creates appropriate identities for developers, testers, and administrators b. Assigns permissions following the principle of least privilege c. Enforces secure authentication d. Prevents developers from accessing or modifying production resources e. Allows administrators full operational control without using the root account for routine tasks.	8	3	3
3		An organization hosts several web applications on Amazon EC2 instances. The security team wants to identify known vulnerabilities in the operating systems and installed software and	7	3	2

		prioritize instances requiring urgent remediation based on vulnerability severity. The organization also wants to automate this assessment instead of relying on manual security checks. As a cloud security engineer, identify and explain the AWS service that can assess, identify, and prioritize vulnerabilities in EC2 workloads, and explain how the findings can be used to improve the organization's security measures [7 Marks]			
4		A bank is moving its online banking application to AWS. Customers must have continuous access to their accounts with minimum downtime. The system should remain available even if a server, database, or Availability Zone fails. It should also be able to tolerate component failures without significantly affecting users. As an AWS architect, design a suitable architecture for this application. Explain how you would ensure availability and tolerate failures using appropriate AWS services, redundancy, multiple Availability Zones, load balancing, automatic recovery, database replication, and backup mechanisms. Explain your design with a simple diagram. [10 Marks]	10	4	3
5		In a scenario of designing an online ticket booking system for a popular concert, the following happens. During normal days, the system handles about 500 users per hour. On the day tickets are released, traffic suddenly spikes to 50,000 users per hour. As an AWS architect, how would you explain the difference between scalability and elasticity in ensuring the system remains available and responsive? Provide a real-world AWS service example for each. [5+2 Marks]	7	4	2
6		A start-up is building a video-streaming platform on AWS. The application must deliver content seamlessly to users across different regions, adapt to changing workloads, and remain cost-effective. Using one of pillars of the AWS Well-Architected Framework, design and architecture that demonstrates how to choose the right resources, monitor performance, and evolve the system over time. Explain how your design ensures both global reach and responsiveness [6+2 Marks]	8	4	4
***** All the best *****					